NAME: _____ Geometry (FL B.E.S.T.)

11.3 Incircle and Circumcircle

Every triangle has exactly one **incenter** (where the three angle bisectors meet, always inside the triangle) and exactly one **circumcenter** (where the three perpendicular bisectors of the sides meet, which can be inside, on, or outside the triangle). The incenter is the center of the **incircle** — the largest circle that fits inside the triangle, tangent to all three sides — with radius $r = \dfrac{\text{Area}}{s}$ (s = semiperimeter). The circumcenter is the center of the **circumcircle** — the unique circle through all three vertices — with radius $R = \dfrac{abc}{4 \cdot \text{Area}}$. This worksheet constructs both centers and computes both radii, including with Heron's Formula for the area.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. The incenter is the intersection of which three lines?

2. The circumcenter is the intersection of which three lines?

3. What is the incircle tangent to?

4. What does the circumcircle pass through?

5. Where is the circumcenter of a right triangle located?

6. Where is the circumcenter of an obtuse triangle located?

7. Where is the circumcenter of an acute triangle located?

8. Is the incenter ever outside the triangle?

9. In an equilateral triangle, how do the incenter and circumcenter compare?

10. A triangle has sides 5, 12, 13. Find its inradius.

11. The same triangle (5, 12, 13) — find its circumradius, and check it two ways.

12. After locating the incenter, what additional measurement is needed to draw the incircle?

13. After locating the circumcenter, what additional measurement is needed to draw the circumcircle?

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14. A triangular plot has sides $7, 15, 20$. Find the radius of the largest circular pond that fits entirely inside it.

15. Prove that $r = \dfrac{\text{Area}}{s}$ for any triangle by connecting the incenter I to all three vertices, splitting the triangle into three smaller triangles.

✓ Answer Key

1. The three angle bisectors
2. The perpendicular bisectors of the three sides
3. All three sides of the triangle
4. All three vertices of the triangle
5. At the midpoint of the hypotenuse
6. Outside the triangle
7. Inside the triangle
8. No — it is always inside
9. They are the same point
10. $r = 2$
11. $R = \dfrac{13}{2}$ (matches half the hypotenuse, since it's a right triangle)
12. The perpendicular distance from the incenter to any one side
13. The distance from the circumcenter to any one vertex
14. $r = 2$
15. Each small triangle has height r (the perpendicular distance from I to that side) and base equal to that side; summing their areas gives $\text{Area} = \dfrac{1}{2}r(a + b + c) = \dfrac{1}{2}r(2s) = rs$, so $r = \dfrac{\text{Area}}{s}$